

Neurons and Bone mechanics

- Suppose that the temporal activity of the space-clamped nerve axon is modelled with the two variable FitzHugh-Nagumo model with an external applied current I_a :

$$\frac{dv}{dt} = f(v) - w + I_a, \quad \frac{dw}{dt} = bv - w, \quad \tau = 1$$

where the function $f(v)$ is the piecewise linear function:

$$f(v) = \begin{cases} -v, & \text{for } v \leq 1 \\ \frac{1}{2}v - \frac{3}{2}, & \text{for } 1 \leq v \leq 5 \\ 6 - v, & \text{for } v \geq 5 \end{cases}$$

and I_a is a positive constant.

Performing an analysis of nullclines, find the range of values of the parameters b and I_a for which the model could have a periodic behaviour.

$$\begin{cases} w = f(v) + I_a \\ w = bv \end{cases}$$

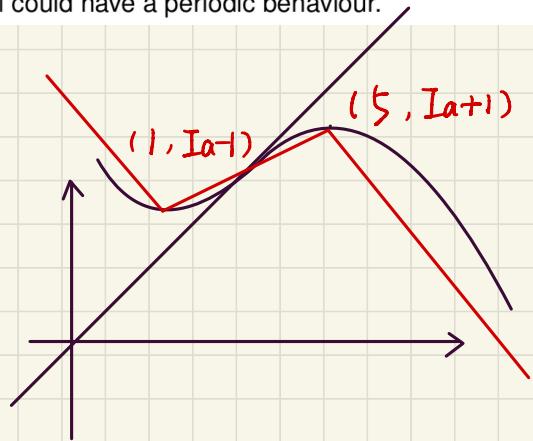
when slope is less and a limit cycle can be observed

$$\begin{cases} \tan \theta = \frac{w_2 - w_1}{v_2 - v_1} < b \end{cases}$$

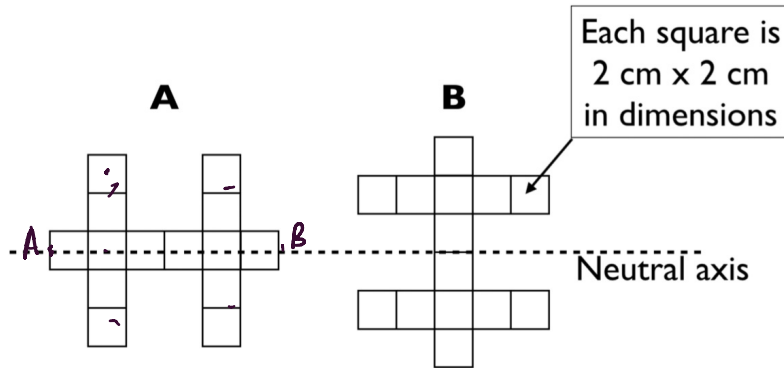
$$I_a - 1 > b$$

$$I_a + 1 < 5b$$

$$\begin{cases} b > 0.5 \\ b + 1 < I_a < 5b - 1 \end{cases}$$



2. Consider two beams *A* and *B* of equal cross-sectional area, with the material distributed around the neutral axis/plane as shown in the figure below. Note that the area of the square used to construct the shapes is equal to $2\text{cm} \times 2\text{cm} = 4\text{cm}^2$. Calculate the area moment of inertia I_A for these two cases. The beam of which shape, *A* or *B*, will resist the bending better? Why?



$$I_A = \int_{-1}^1 y^2 \times 2 dy + 4 \int_1^5 y^2 \times 2 dy = \frac{1016}{3} \text{ cm}^2$$

$$I_B = \int_{-6}^6 y^2 \times 2 dy + 4 \int_2^4 y^2 \times 4 dy = \frac{1760}{3} \text{ cm}^2$$

$$I_B > I_A$$

Bone B is better